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SkinNexus® Collag3n

Advancing the future of Collagen Banking

Global Except US

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SUMMARY FILE

SkinNexus® Collag3n C	BC10239
SkinNexus® Collag3n P	BC10252
Origin - Description	100% human identical recombinant vegan collagen III
Regulatory data	
	• **SkinNexus Collag3n C**
INCI	Mannitol (and) Collagen
China	All components are listed in Inventory of Existing Cosmetic Ingredient in China (IECIC 2021)
CAS#	69-65-8; 9007-34-5
	• **SkinNexus Collag3n P**
INCI	Mannitol (and) sh-Polypeptide-123
China	sh-Polypeptide-123 is NOT listed in Inventory of Existing Cosmetic Ingredient in China (IECIC)
CAS#	69-65-8; 3121591-49-6
Appearance	White to off white homogeneous powder
Natural labels	None
Naturalness content (ISO 16128)	100% from natural origin
Cosmetic use	
Properties	• Supports fibroblasts and keratinocytes proliferation, contributing to skin regeneration across dermal and epidermal layers (*in vitro*) • Supports synthesis of Collagen I, III and V, contributing to dermal matrix architecture (*in vitro*) • Supports epidermal thickness and regeneration (*in vitro*) • Helps reduce the appearance of wrinkles (*in vivo*) • Helps improve lower face skin tonicity and sagging (*in vivo*)
Applications	• Preventive anti-aging skincare • Anti-wrinkle and anti-sagging facial care, facial contour redefining • Collagen banking skincare concepts • Pre- and post-procedure skincare
Formulation data	
Concentration of use	1%
Solubility	Soluble in water, insoluble in oils and fats
Incorporation method	Incorporate into the cosmetic product up to 75°C or at room temperature for cold processing
Optimal pH	4-8



The reality of collagen decline: why collagen banking matters

Collagen is the predominant structural protein of human skin, representing approximately 75% of the dry weight of the dermis and forming the core architectural framework of the extracellular matrix [Oikarinen, 1994; Salamito *et al*, 2025]. Through its highly organized fibrillar network, collagen plays a central role in conferring **mechanical strength, elasticity, and structural integrity** to the skin, thereby ensuring its resistance to mechanical stress and its ability to recover from deformation [Salamito *et al*, 2025; Jadach *et al*, 2024]. Importantly, dermal collagen is not a static component: it is continuously renewed through tightly regulated processes of **synthesis, assembly**, and **degradation** that collectively maintain extracellular matrix homeostasis throughout life.

This balance begins to shift relatively early, from around the mid-twenties, well before the onset of clinically visible signs of skin aging [He *et al*, 2023; Reilly *et al*, 2021]. Early age-related alterations in the dermal collagen network include a progressive reduction in collagen density, a loss of fibrillar organization, and a measurable decline in the biomechanical performance of the dermis [He *et al*, 2023]. These initial changes reflect subtle but cumulative impairments in matrix renewal which will precede macroscopic manifestations such as wrinkles, sagging, and loss of firmness.

From early adulthood onward, **dermal collagen content decreases** at an estimated rate of **approximately 1% per year**, indicating a chronic imbalance between collagen synthesis and degradation processes [Varani *et al*, 2006; Reilly *et al*, 2021]. This progressive decline is primarily driven by **age dependent fibroblast dysfunction**, characterized by reduced collagen synthetic activity, diminished responsiveness to mechanical stimuli, and impaired mechano-transduction within the extracellular matrix. In parallel, the dysregulation of matrix remodeling pathways further compromises collagen renewal and accelerates dermal matrix weakening. Together, these mechanisms contribute to dermal thinning, decreased firmness and sagging, and reduced regenerative capacity, which are well-established biological hallmarks of **skin aging** [Naharro-Rodriguez *et al*, 2025; Quan *et al*, 2023].

These scientific findings have driven a paradigm shift from late corrective strategies toward preventive approaches focused on preserving the skin's **collagen capital**, summarized under the "**collagen banking**" concept. This concept aims to maintain a structurally and biologically optimal and human-like collagen environment over time, to sustain dermal integrity, biomechanical resilience, and long-term functional performance before irreversible collagen depletion occurs.

By acting early and sustainably, collagen banking represents a forward-looking strategy to preserve skin quality, resilience and youthfulness across the lifespan.

Collagen III: a key target for collagen banking

Type III collagen is a major fibrillar collagen of the dermal extracellular matrix, encoded by the COL3A1 gene.
In humans, each α1(III) chain contains approximately **1.466 amino acids**, and three identical chains assemble into the mature triple-helical collagen III molecule. Functionally, **collagen III plays a key role in skin elasticity, mechanical flexibility** (the ability of the dermal matrix to adapt to mechanical stress), **and tissue repair capacity** [Kuivaniemi & Tromp, 2019; Kanniyappan *et al*, 2025].

Collagen III is also commonly referred to as "**baby collagen**" because of its **higher relative abundance during early life**, where it represents **approximately 18–21% of total dermal collagen in fetal skin**, compared with only 8–11% in adult skin [Smith *et al*, 1986; Reilly *et al*, 2021].

Importantly, **collagen III declines faster than collagen I with age**, leading to a progressive imbalance in the collagen network and an increase in the collagen I/III ratio. These age related shifts are associated with changes in fibril organization, increased matrix stiffness, and a gradual loss of dermal flexibility and regenerative potential [Cheng *et al*, 2011; Gao *et al*, 2022]. As these alterations occur early in the skin aging process, targeting collagen III aligns particularly well with the concept of collagen banking, which focuses on preserving the quality and functionality of the dermal matrix before irreversible structural degradation takes place.

Beyond aging, collagen III is recognized as the **predominant fibrillar collagen synthesized during the early wound healing phase**, where it forms a provisional and highly organized scaffold that supports fibroblast migration, cell-matrix interactions, and efficient extracellular matrix remodeling [Singh *et al*, 2023; Mathew-Steiner, 2021].

Supplying Collagen III to the skin provides a key approach to supporting dermal youthfulness and regenerative potential within a proactive collagen banking strategy.

Recombinant type III collagen: the next-gen solution for collagen banking

For decades, animal-derived collagens have been widely used in cosmetic formulations. While they offer basic moisturizing or film-forming benefits, these materials inherently suffer from biological variability, batch-to-batch inconsistency, and limited reproducibility [Guo *et al*, 2023]. In addition, animal collagens raise ethical concerns, are increasingly misaligned with clean beauty expectations and carry potential contamination risks, including residual pathogens or immunogenic fragments [Lynn *et al*, 2004].

In response to these limitations, recombinant collagen technologies were developed to **improve purity, reproducibility and process control**. Compared with animal-derived collagen, recombinant human collagen has been reported to also offer improved biological activities, highlighting its strong potential for cosmetic applications [Jie *et al*, 2025; He *et al*, 2025]. These attributes have positioned **recombinant collagen** as a **promising alternative** for **next-generation, non-animal collagen** solutions.

However, early or first generation recombinant collagens have only partially fulfilled this promise. Plant-derived or first generation recombinant collagens often fail to fully replicate the molecular identity of native human collagen or rely on fusion tags and sequence modifications that can alter biological behavior [Bai *et al*, 2023]. Moreover, despite their technological advantages, these first generation recombinant collagens remain associated with high production costs, which continue to limit scalability and broad industrial adoption [Jie *et al*, 2025].

As a result, existing recombinant approaches have struggled to simultaneously deliver 100% identity to native collagen, biological efficacy, and economic feasibility, leaving a clear innovation gap within the collagen market.

SkinNexus Collag3n: closing the innovation gap

SkinNexus Collag3n, 100% human-identical recombinant type III collagen, is the best in class biotechnology solution for high-performance collagen banking.

An AI-driven selection process of SkinNexus Collag3n

To identify SkinNexus Collag3n, an advanced multi-step selection strategy was implemented combining artificial intelligence, bioinformatics and experimental validation (Fig. 1).

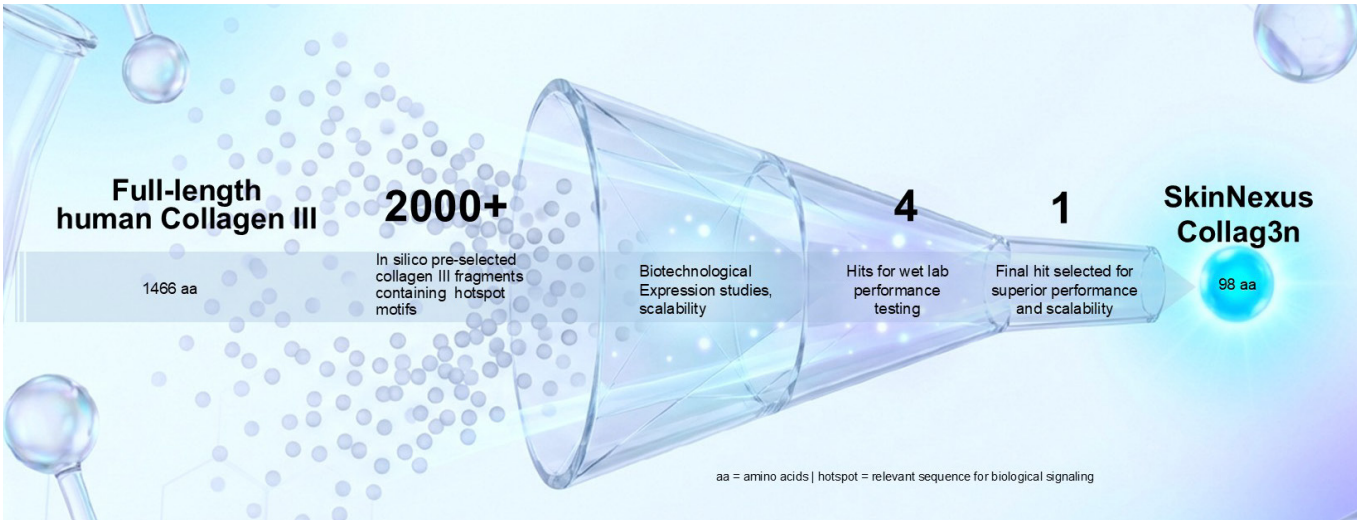


Figure 1 - Schematic illustration of the AI-driven selection process.

The process started with an **AI-driven in silico hotspot analysis** of the full length human collagen III molecule. In this context, hotspots are defined as relevant amino acid sequences involved in biological signaling and key cell-matrix interactions. More than **2,000 collagen III peptide scaffolds** containing such biologically relevant hotspot motifs were identified based on a predicted involvement in skin related biological functions.

In a second step, candidates were filtered according to expression capacity and **industrial scalability** to ensure compatibility with large scale recombinant production. This manufacturing oriented selection step enabled the early elimination of sequences unable to meet robustness, yield and industrial performance requirements.

From this refined dataset, **4 leading candidates** were selected and subjected to **wet-lab biological screening**. This experimental validation phase enabled direct comparison of biological relevance across candidates.

Among these fragments, **one sequence** clearly distinguished itself, **combining superior biological performance with robust scalability**. This fragment was ultimately selected as **SkinNexus Collag3n**, a recombinant collagen III fragment designed to meet both biological efficacy and industrial manufacturing requirements.

This AI-guided and data-driven approach illustrates how the convergence of **biological discovery intelligence** and **biomanufacturing process expertise** enabled the identification of a next generation collagen solution optimized for **high-performance collagen banking**.

Characteristics of SkinNexus Collag3n

Number of amino acids: 98

Molecular weight: 8802 g/mol

Description: 100% Human-identical recombinant collagen III with dual integrin-binding hotspots.

SkinNexus Collag3n sequence contains integrin binding motifs that may support biologically relevant cell-matrix signaling. A key feature is the presence of a GETGER motif, belonging to the GxxGER integrin binding motif family. Structural studies have shown that collagen-binding integrins such as $\alpha 2\beta 1$ recognize the GER triplet, with the glutamate residue coordinating the MIDAS metal ion in the integrin domain [Emsley *et al*, 2000; Knight *et al*, 2000]. Sequence variability within GxxGER motifs is tolerated as long as the GER core is preserved [Siljander *et al*, 2004; Hamaia et Farndale, 2014]. In addition, SkinNexus Collag3n contains an RGD motif, a canonical integrin-binding sequence. While RGD sites are cryptic (hidden) in native fibrillar collagen, several studies have shown that partial unfolding or denaturation of collagen matrices leads to the functional exposure of RGD motifs. These exposed sites can engage integrins such as $\alpha v\beta 3$ and $\alpha 5\beta 1$, thereby promoting cell adhesion, matrix remodeling and cellular differentiation [Davis, 1992; Taubenberger *et al*, 2010; Davidenko *et al*, 2016].

SkinNexus Collag3n sequence is 100% identical to human collagen III [Collagen α-1(III) chain – Homo Sapiens (Human)]. It does not contain any fusion tags (such as His-Tag, INCI Hexapeptide-40) or additional amino acids or sequence modification.

Biotechnological process

SkinNexus Collag3n is manufactured through a sustainable and clean biotechnology process based on fermentation (Fig. 2). The process begins with targeted strain engineering, using *Pichia pastoris* as a tailored microbial factory designed to express the human identical collagen III polypeptide. Controlled fermentation ensures consistent expression, purity and scalability under tightly monitored conditions. After yeast removal by frame filtration, the collagen III polypeptide secreted in the fermentation broth is then isolated and purified by ultrafiltration and exchange-ion chromatography. This clean bio-manufacturing approach delivers **pure, vegan, and safe recombinant collagen III**, compatible with large scale industrial production.

Solubility: water soluble.

Heat stability: stable in water-based solution after 5 hours of heating at 75°C. It can be incorporated in cosmetic formulations up to 75°C or at room temperature for cold processing.

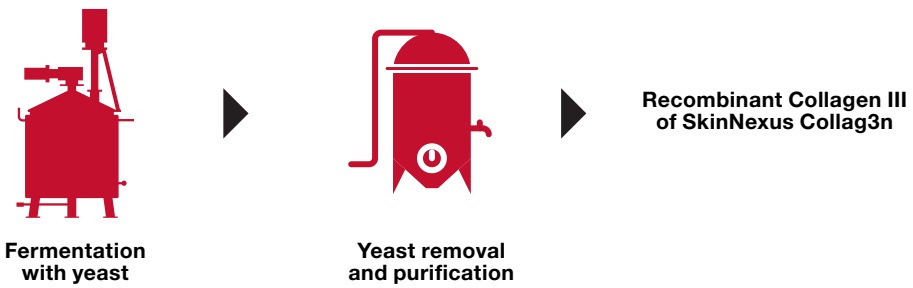


Figure 2 - Schematic biotechnological process flow of recombinant Collagen III of SkinNexus Collag3n.

Superior performance of SkinNexus Collag3n *vs* competitor benchmarks

The ability of the recombinant Collagen III from SkinNexus Collag3n to stimulate type I collagen synthesis in human dermal fibroblasts was evaluated in comparison with 4 competitor recombinant collagen benchmarks, differing in type (I, III and XVII) and molecular weight (from 4.9 kDa to 43.6 kDa):

- Benchmark T1, a leading market benchmark recombinant collagen III (INCI: Sr-(sh polypeptide 69-dipeptide 19); MW 4.9 kDa), also evaluated in our clinical study *vs* SkinNexus Collag3n cf page 17),
- Benchmark T2, a higher molecular weight recombinant collagen III (INCI: sh-polypeptide-123; MW 43.6 kDa),
- Benchmark T3, a recombinant collagen XVII (INCI: collagen amino acids; MW 5.8 kDa),
- Benchmark L, a recombinant collagen I (INCI: Nicotiana Benthamiana Hexapeptide-40 sh-Polypeptide-47; MW 10.7 kDa).

All products were tested at the same concentration (0.04 ppm of the recombinant polypeptide), except for Benchmark L, which was evaluated at its recommended use concentration (0.02 ppm of the recombinant polypeptide).

We observed that treatment with the recombinant Collagen III from SkinNexus Collag3n induced a significant +46% stimulation of type I collagen production (146%, SEM ± 3%, $p<0.001$) compared with the untreated control (normalized at 100%, SEM ± 9%) (Fig. 3). Notably, the recombinant collagen III from SkinNexus Collag3n was between 2-fold and 4-fold more effective ($p<0.01$ or $p<0.001$ depending on the product) than the competitor benchmarks. Neither the leading benchmark T1, nor the benchmarks T3 and L, demonstrated a significant effect on collagen I stimulation under the same experimental conditions. Only the benchmark T2 induced a limited +23% increase in collagen I synthesis (123%, SEM ± 4%, $p<0.05$).

Overall, these results demonstrate the **superior efficacy** of the recombinant collagen III from SkinNexus Collag3n compared with competitor recombinant collagen based benchmarks, highlighting its strong potential to reactivate dermal matrix synthesis **at very low concentrations**.

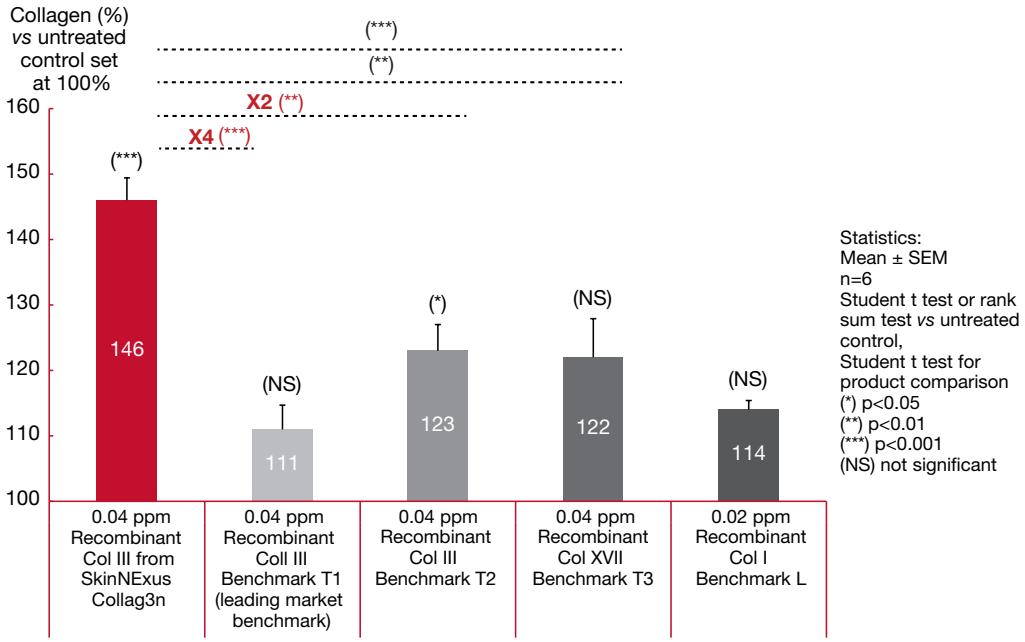


Figure 3 - Comparison of SkinNexus Collag3n with recombinant collagen benchmarks on collagen I synthesis by human dermal fibroblasts.

MATERIALS&METHODS

Study number
EMB201201-2024-151-IDCD

Tested products
Peptide content in all samples was determined by nitrogen quantification (Kjeldahl method), allowing accurate normalization and dilution of the tested products.
The recombinant collagen III from SkinNexus Collag3n (BTE_0000951_001), Benchmark T1 (MIS_0000272_001), Benchmark T2 (MIS_0000271_001), and Benchmark T3 (MIS_0000286_001) were diluted in water and tested at a final concentration of 0.04 ppm (0.04 µg/ml) of polypeptide. Benchmark L (E250130.01) was tested at its maximum recommended use level, corresponding to 0.02 ppm (0.02 µg/ml) of polypeptide.

Cell Culture and Treatment
Normal human fibroblasts derived from a 19-year-old female donor were seeded in 96-well plates and cultured for 4 days at 37°C and 5% CO_2. Cells were treated with the tested products at a final concentration of 0.04 or 0.02 ppm of polypeptide. An untreated condition with only culture medium was used to measure basal collagen I production. The level of collagen I was measured after 48 hours of incubation at 37°C and 5% CO_2.

Quantification of Collagen I
A lysis reagent was then added to measure DNA content without solubilizing collagen I deposited at the bottom of the wells. DNA quantity was measured using Quant-iT PicoGreen reagents (ThermoFisher Scientific), and collagen I was quantified using DELFIA TRF technology (Revvity).

Results and statistics
Fluorescence values were normalized to DNA content to obtain a stimulation ratio for each well. The percentage of stimulation of collagen I synthesis was then calculated relative to the untreated control and expressed as mean percentage ± SEM (standard error of the mean) with n=6. Statistical analysis was performed using SigmaPlot software (version 12.5) (Systat Software Inc. San Jose, California United States of America), with a significance threshold set at 5% ($p<0.05$).

SkinNexus Collag3n is a next-generation 100% human-identical recombinant collagen III for collagen banking, optimally conceived through an AI-guided process integrating biological relevance with industrial feasibility. As demonstrated *in vitro*, it has superior performance compared with market recombinant collagen benchmarks.

Safety / tolerability of the product

SkinNexus Collag3n was evaluated to ensure its safety in the recommended conditions of use.

SkinNexus Collag3n is available as a preservative free powder containing ~5000 ppm of recombinant Collagen III formulated with mannitol as an excipient.

SkinNexus Collag3n C INCI name: Mannitol (and) Collagen

SkinNexus Collag3n P INCI name: Mannitol (and) sh-Polypeptide-123

Naturalness content (according to ISO 16128): 100% from natural origin

Recommended dose of use: 1%

Description



Example: illustration of a commercial sample of SkinNexus Collag3n incorporated into a face emulsion and an aqueous solution at 1%.



Demonstrated efficacy

SkinNexus Collag3n supports collagen banking by reinforcing the skin's collagen capital through powerful *in vitro* proven activity in the dermis combined with benefits at the epidermal level. Two clinical studies validated its performance on visible signs of skin aging, *vs* a leading market benchmark and *vs* placebo.

In vitro, the biological activity of SkinNexus Collag3n was demonstrated.

At the dermal level, with:
- promotion of fibroblast proliferation,
- stimulation of collagen I, III and V synthesis in a 3D-reconstructed dermis model.

At the epidermal level, with:
- enhanced keratinocyte proliferation and adhesion,
- increased epidermal thickness in human skin explants.

In vivo, two clinical studies demonstrated the cosmetic benefits of SkinNexus Collag3n.

Compared to a leading benchmark, the first clinical study showed superior efficacy on wrinkle reduction and skin sagging improvement. These results were further substantiated in a second placebo-controlled clinical study further confirming rapid and sustained benefits on wrinkles and skin tonicity.

EFFICACY

Dermal level: promotion of fibroblast proliferation

in vitro

OBJECTIVE

Structural skin aging is driven by progressive alterations of the dermal extracellular matrix, notably collagen degradation and reduced collagen synthesis, leading to wrinkles and skin laxity. Aging related changes are closely associated with impaired dermal fibroblast function and reduced fibroblast density. A reduction in the total number of dermal fibroblasts in aged subjects compared with young individuals was reported, highlighting the contribution of fibroblast depletion to age related dermal deterioration [Varani *et al*, 2006; Fisher *et al*, 2008; Zhang *et al*, 2024; Zorina *et al*, 2022].

As fibroblasts are the primary cells responsible for collagen synthesis, maintaining their number is considered a factor in preserving long-term dermal collagen homeostasis, a concept often referred to as collagen banking. In this context, fibroblast proliferation represents a relevant *in vitro* readout to assess an ingredient's ability to support the maintenance of these key dermal cells involved in structural skin aging.

The aim of the present study was to evaluate SkinNexus Collag3n's capacity to promote human dermal fibroblast proliferation in a monolayer model.

RESULTS & DISCUSSION

Treatment with SkinNexus Collag3n at 0.004% resulted in a significant increase in fibroblast proliferation (+33%, SD ± 14%, p<0.001) compared with the untreated control normalized at 100% (SD ± 8%) (Fig. 4).

CONCLUSION

By supporting the *in vitro* proliferation of dermal fibroblasts, SkinNexus Collag3n may contribute to mechanisms associated with collagen banking, aiming to help preserve the skin's collagen capital over time.

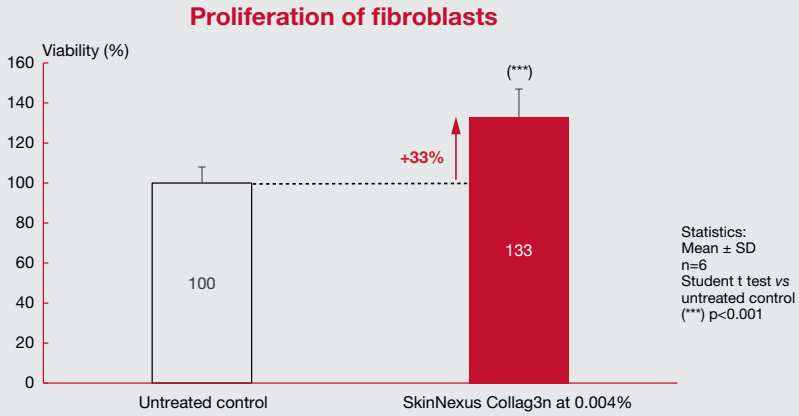


Figure 4 - Fibroblast proliferation assay on human fibroblasts.

MATERIALS&METHODS

Study number

EMB201201-2023-886-IDCD

Tested products

Recombinant collagen III from SkinNexus Collag3n (BTE_0000951_001), at 0.2 ppm (0.2 µg/ml) (corresponding to 0.004% of SkinNexus Collag3n).

Cell culture and treatment

Abdomen normal human fibroblasts from a donor of 26-year-old were seeded in 96-well plates with Dulbecco's Modified Eagle Medium (DMEM) supplemented with calf serum at a concentration of about 2%, and antibiotics. Just after the seeding, cells are treated with the recombinant Collagen III from SkinNexus Collag3n diluted in culture medium. Cells cultured with culture medium alone served as the untreated condition for assessing basal fibroblast proliferation. Cells were then incubated at 37°C, 5% CO_2 for 4 days.

Viability quantification

Cell viability was assessed using the MTT assay (3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide). Culture medium was first removed, and 200 µl of MTT solution (0.5 mg/ml in PBS) was added to each well. Plates were incubated for 3 hours at 37°C in a humidified atmosphere with 5% CO_2 under gentle agitation. Following incubation, the resulting formazan crystals, produced by the metabolic activity of viable cells, were solubilized using DMSO (Dimethyl Sulfoxide). Absorbance was then measured at 570 nm using a multiplate reader (EnVision 2104 Multilabel Reader, PerkinElmer).

The amount of formazan formed is directly proportional to cellular metabolic activity and therefore reflects cell viability, which is commonly used as an indirect measure of cell number in proliferation assays.

Results and statistics

The percentage of fibroblast proliferation stimulation was determined by normalizing the absorbance value of the tested ingredient to the untreated control, set at 100%.

Results are presented as mean percentage ± SD (standard deviation). Statistical analysis of data was performed using SigmaPlot software (V12.5) (Systat Software Inc. San Jose, California United States of America), with n=6. Statistical analysis has been assessed by a Student t test according to software recommendation with threshold of significance set to 5% (p<0.05).

EFFICACY

Dermal level: boosted collagen production by fibroblasts

in vitro

OBJECTIVE

Dermal collagen homeostasis is known to shift relatively early in life, from around the mid twenties, well before visible signs of skin aging emerge [He *et al*, 2023; Reilly *et al*, 2021]. From early adulthood onward, total dermal collagen content declines at an average rate of approximately 1% per year, reflecting a persistent imbalance between collagen synthesis and degradation [Varani *et al*, 2006; Reilly *et al*, 2021].

Among the collagens present in skin, collagens I, III and V play complementary and interdependent roles within the fibrillar dermal matrix. Collagen I provides tensile strength, while collagen III is essential for fibril elasticity and tissue compliance and is known to decline more quickly with age, contributing to early loss of elasticity potential [Chen *et al*, 2011; Gao *et al*, 2022]. Collagen V, although quantitatively minor, plays a key regulatory role by guiding the assembly and size of collagen fibrils through its interaction with collagen I [Sun *et al*, 2011; Wenstrup *et al*, 2004].

Supporting the early and sustained production of fibrillar collagens I, III and V represents a biologically relevant approach for collagen banking, i.e. to preserve dermal architecture before irreversible structural decline occurs.

In this context, the aim of the present study was to assess SkinNexus Collag3n's capacity to enhance the synthesis of collagen types I, III and V by dermal fibroblasts using a 3D reconstructed human dermis model prepared from cells of a young donor (18 years old).

RESULTS & DISCUSSION

SkinNexus Collag3n at 0.004% significantly stimulated collagen production by human fibroblasts in a 3D dermis model (Fig. 5 & 6). Relative to their respective untreated controls, collagen I, III and V synthesis increased by +48% (SEM ± 17%, $p<0.1$), +82% (SEM ± 21%, $p<0.01$), and +71% (SEM ± 22%, $p<0.01$), respectively.

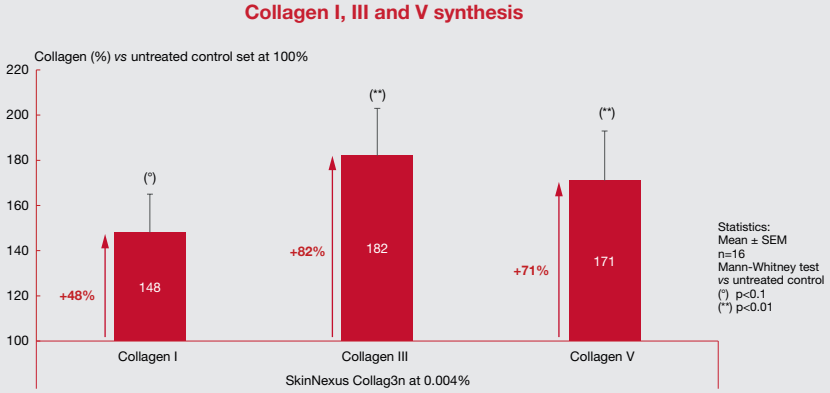


Figure 5 - Collagen I, III and V synthesis in a reconstructed dermis model.

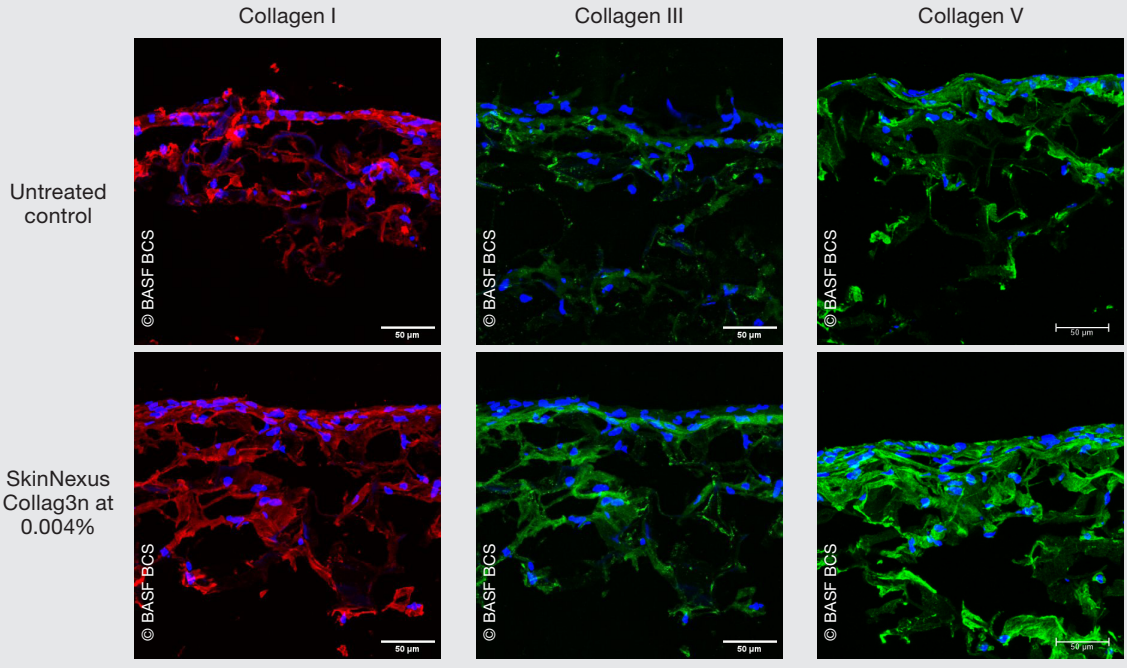


Figure 6 - Illustrative pictures of Collagen I, collagen III and collagen V immunostaining in a reconstructed dermis model (in red: collagen I staining, in green: collagen III and collagen V staining; in blue: DAPI stained nuclei).

in vitro

CONCLUSION

In vitro, on 3D reconstructed human dermis, SkinNexus Collag3n stimulates the synthesis of key structural collagens, with a particularly strong effect on collagen III. This is especially relevant, given that collagen III declines early and more rapidly with age, a process that contributes to the progressive loss of fibril organization. SkinNexus Collag3n may thus support collagen banking to maintain dermal structure and resilience.

MATERIALS&METHODS

Study number

EMB201201-2024-028-MEXP

Tested products

Recombinant collagen III from SkinNexus Collag3n (BTE_0000951_001), at 0.2 ppm (0.2 µg/ml) (corresponding to 0.004% of SkinNexus Collag3n).

3D dermis culture and treatment

Human reconstructed dermis models (Mimederm) were obtained by seeding fibroblasts isolated from a young adult female donor (18 years old) onto an acellular matrix composed of collagen, chitosan and glucosaminoglycan (Mimedisc). The constructs were cultured in a specific cell-growth medium consisting of DMEM (Dulbecco's Modified Eagle Medium)/F12 supplemented with 10% FBS and antibiotics. Two conditions were tested with 4 reconstructed dermis per condition: the untreated control and the SkinNexus Collag3n treatment (diluted in PBS), applied in the culture medium every two days from day 7 to day 28. The experiment stopped on day 28.

Collagen I, collagen III and collagen V analysis

The expression of collagen I, III and V were evaluated by immunostaining followed by image based quantification. Tissue cryosections (10 µm) were fixed in acetone at -20°C for 10 minutes, then incubated with 2% serum for 2 hours at room temperature. After washing, the sections were incubated for 2 hours at room temperature with anti collagen I antibodies (Invitrogen), anti collagen III antibodies (Novotec) and anti-collagen V antibodies (Clinisciences). Secondary antibodies (ThermoFisher) were then applied. Finally, the slides were mounted with an antifade medium (Vectashield) containing DAPI (4',6-diamidino-2-phenylindole) for nuclear staining, and analyzed by fluorescence microscopy (LEICA SP8).

Results and statistics

The expression of collagen I, collagen III and collagen V was quantified by image analysis using FIJI software (ImageJ2, open source). Results are presented as mean ± SEM (standard error of the mean). A total of 16 images were acquired per condition (4 reconstructed dermis per condition and 4 images per reconstructed dermis, n=16).

Statistical analysis was performed using SigmaPlot software (V12.5) (Systat Software Inc. San Jose, California United States of America). A two tailed test (Mann Whitney) was used to compare treated samples with untreated control according to software recommendation. The significance threshold was set at 5% ($p<0.05$).

EFFICACY

Epidermal level: boosted proliferation and adhesion of keratinocytes

in vitro

OBJECTIVE

Beyond its structural role in the dermis, collagen contributes to the regulation of epidermal homeostasis and regenerative processes. Indeed, dermal extracellular matrices enriched in fibrillar collagens provide essential biochemical and biophysical cues that support keratinocyte adhesion and proliferation, highlighting the importance of early cell-matrix interactions in epidermal renewal [Wong *et al*, 2025; Fujisaki *et al*, 2019]. Some recombinant collagen type III have also been reported to enhance keratinocyte adhesion and proliferation *in vitro*, suggesting that they may support epidermal renewal [Wang *et al*, 2026; Kim *et al*, 2024].
By sustaining keratinocyte anchorage and proliferative capacity over time, collagen III-based strategies may help preserve epidermal renewal potential, supporting collagen banking approaches aimed at maintaining long-term epidermal functionality.
The aim of the present study was therefore to evaluate the ability of SkinNexus Collag3n to promote human epidermal keratinocyte cell adhesion and proliferation in a monolayer model.

RESULTS & DISCUSSION

SkinNexus Collag3n at 0.004% significantly enhanced both keratinocyte proliferation and adhesion (Fig. 7). It increased keratinocyte proliferation by +31% (SEM ± 5%, p<0.001) and keratinocyte adhesion by +74% (SEM ± 13%, p<0.001) relatively to the untreated control normalized at 100% (SEM ± 2%).

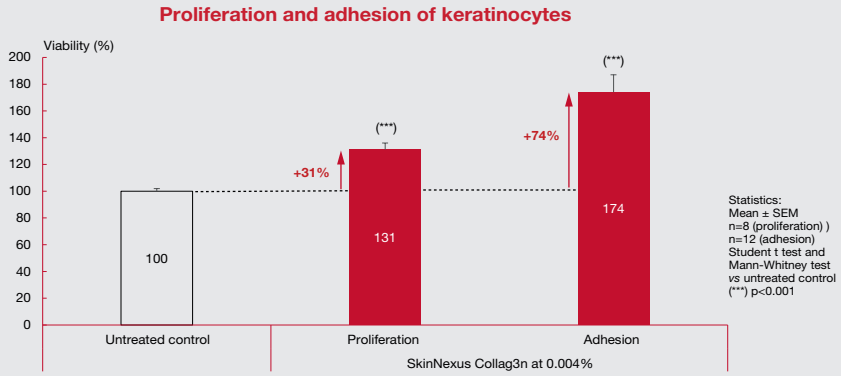


Figure 7 - Proliferation and adhesion of human epidermal keratinocytes.

CONCLUSION

In vitro, SkinNexus Collag3n enhanced epidermal keratinocyte adhesion and proliferation. It may thus contribute to mechanisms associated with collagen banking, aiming to maintain epidermal functionality and renewal over time.

MATERIALS&METHODS

Study number
EMB201201-2024-030-CC (Proliferation)
EMB201201-2024-029-CC (Adhesion)

Tested products
Recombinant collagen III from SkinNexus Collag3n (BTE_0000951_001), at 0.2 ppm (0.2 µg/ml) (corresponding to 0.004% of SkinNexus Collag3n).

Proliferation test
Normal human epidermal keratinocytes (NHEK), obtained from 37- and 40-year-old healthy donors, were seeded in 48-well plates at a rate of 50,000 cells/cm² and cultured in a specific culture medium DMEM (Dulbecco's Modified Eagle Medium) mixed with Ham's F-12 medium and supplemented with fetal calf serum and an antibiotic, for 6 hours at 37°C, under 5% CO_2. Tested product was evaluated on the NHEK from 2 donors with 4 replicates per donor. NHEK were incubated for 24 hours with tested product in DMEM supplemented with a penicillin–streptomycin blend (Gibco, Fischer Scientific). An untreated condition corresponding to NHEK treated with only culture medium was used to measure the basal proliferation.
Cell proliferation was then assessed using the colorimetric MTT assay (viability test).

Adhesion test
NHEK obtained from 3 donors aged 37, 40, and 52 years were incubated in suspension at a density of 50,000 cells/cm² with the tested product for 1 hour at 37°C under 5% CO_2. An untreated condition, in which cells were exposed to the same culture medium without test product, was used to determine basal adhesion levels. Each condition was evaluated using the NHEK from the 3 donors with 4 replicates per donor.
Then, these cell suspensions were seeded into 48 well plates and allowed to adhere for 20 minutes. After a washing step, adherent cells were further incubated for 5 hours in fresh culture medium without test product at 37°C under 5% CO_2.
The number of adherent cells was measured thanks to the colorimetric MTT assay (viability test).

Viability test (MTT test)
Cell viability was assessed using the MTT assay. After a washing step, cells were incubated with MTT at 0.25 mg/ml for 2h30 at 37°C. Then, the cells were washed and incubated with 1ml of DMSO. The absorbance was read at 540 nm with a multimode microplate reader (Varioskan Flash).

Results and statistics
Results are expressed as mean values ± SEM (standard error of the mean), with n=8 for the proliferation assay and n=12 for the adhesion assay, relative to the untreated control normalized at 100%. Statistical analyses were performed using SigmaPlot software (version 12.5, Systat Software Inc.). Depending on data distribution, comparisons between treated samples and untreated controls were carried out using either a Student's t test or a two tailed Mann–Whitney test according to the software recommendation. The significance threshold was set at 5% (p<0.05).

EFFICACY

Epidermal level: epidermal regeneration booster

ex vivo

OBJECTIVE

Among fibrillar collagens, collagen III plays a crucial role in wound healing, as it predominates during the early repair phase and supports re-epithelialization and tissue remodeling [Singh *et al*, 2023; Mathew-Steiner *et al*, 2021].
Preserving a collagen III-rich microenvironment is therefore considered key for maintaining epidermal regenerative capacity, in line with collagen banking strategies aimed at sustaining long-term skin renewal. Building on this understanding, *ex vivo* wound healing models based on biopsies offer a relevant approach to assess the regenerative efficacy of active ingredients as the growth of the neo-epidermis provides a direct indicator of epidermal renewal [Wang *et al*, 2019; Choudhary *et al*, 2024].
In this context, the aim of the present study was to evaluate SkinNexus Collag3n's ability to enhance neo-epidermis thickness growth in a human skin explant wound healing model.

RESULTS & DISCUSSION

Neo-epidermis average thickness measurement

This study showed a significantly greater increase in neo-epidermis average thickness when treated topically with SkinNexus Collag3n at 1% (Fig. 8 and 9). 8 days post-lesion, SkinNexus Collag3n at 1% resulted in a +40% increase of neo-epidermis average thickness compared to the untreated control (63 µm ± SEM 5 µm *vs* 45 µm ± SEM 4 µm, $p<0.05$).

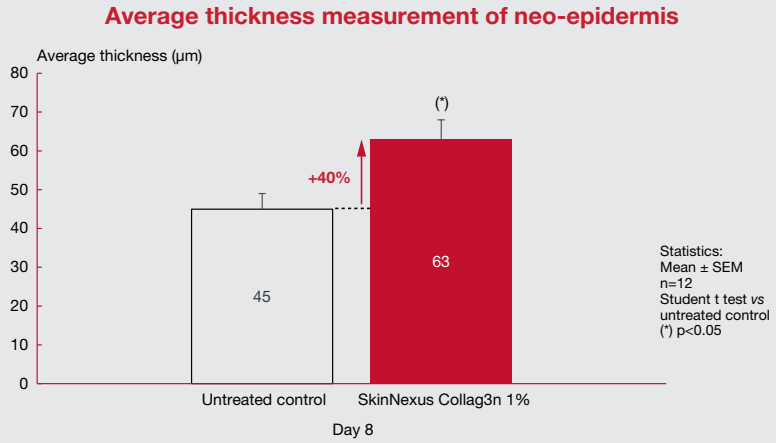


Figure 8 - Neo-epidermis average thickness measurement in wounded skin biopsies.

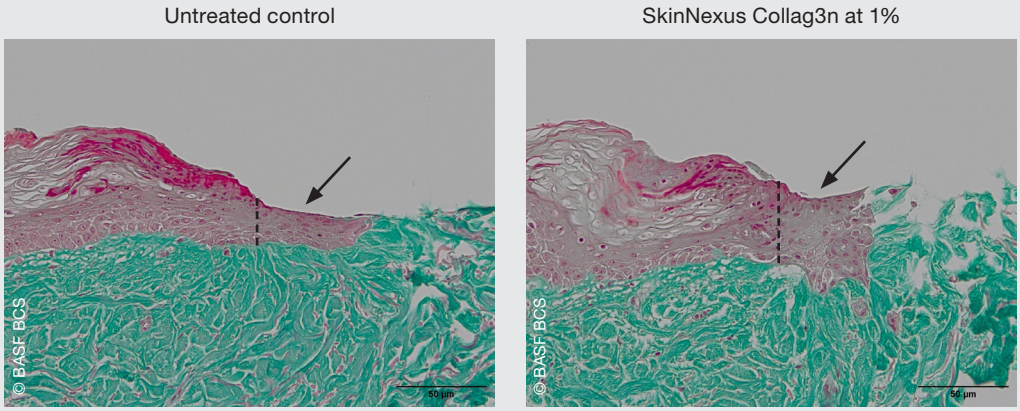


Figure 9 - Illustrative pictures of neo-epidermis average thickness progress at day 8 post-lesion (after Masson's trichrome staining).

CONCLUSION

Ex vivo, topical application of SkinNexus Collag3n at 1% on injured skin explants enhanced the average thickness of the newly regenerated epidermis. SkinNexus Collag3n may thus contribute to mechanisms associated with collagen banking, aiming to maintain epidermal functionality, renewal and skin resilience over time.

MATERIALS&METHODS

Study number
EMB201201-2024-131-HI

Tested products
Recombinant collagen III from SkinNexus Collag3n (MOL_0001121_001), at 50 ppm (50 µg/ml) (corresponding to 1% of SkinNexus Collag3n) diluted in Dulbecco's phosphate-buffered saline solution (DPBS).

Treatment
Rectangular explants of 10x15 mm (1.5 cm^2) were prepared from abdominoplasty tissue obtained from a 49-year-old woman. The explants were maintained in an appropriate culture medium for explant survival at 37°C in a humid atmosphere with 5% CO_2.
At Day 0 (start), two lesions per explant were created using a 3.5 mm biopsy punch at a depth sufficient to remove the epidermis and the superficial part of the dermis. Lesions were made immediately before product application. At Day 0 (after lesion), Day 2, Day 5, and Day 7, the SkinNexus Collag3n solution was applied topically at 3 µl per "wounded" explant of 1.5 cm^2 (2 mg/cm^2) and spread with a small spatula (3 explants per condition). The culture medium was half-renewed on days 2, 5, and 7. Injured explants without product solution application, referred to as untreated controls, served as controls for analysis on day 8. On days 0 and 8, explants were collected and placed in a buffered formaldehyde solution. After 24h, samples were dehydrated and embedded in paraffin. Sections of 5 µm thickness were prepared using a microtome and mounted on glass histology slides.

Neo-epidermis average thickness measurement method
As the neo-epidermis is not uniformly thick, an average thickness was calculated using the surface area to length ratio, allowing local thickness variations to be taken into account. The area and length of the newly formed neo-epidermis were measured on 5 µm paraffin embedded human skin sections after Masson's trichrome staining. Observations and image acquisitions were performed using a Leica DMLB microscope equipped with an Olympus DP digital imaging system (ImageJ software).

Results and statistics
The relative average thickness of the neo-formed epidermis expressed in µm was calculated as the ratio between the surface area (A) of the neo-epidermis (in $µm^2$) and its length (L) (in µm), according to the following formula:
Average thickness (µm) = A (neo epidermis area in $µm^2$) / L (neo epidermis length in µm)
Percentage variation relative to the untreated control was then calculated at day 8 post lesion (D8).
Results are presented as mean ± SEM (standard error of the mean). Statistical analysis was performed using SigmaPlot software (version 12.5; Systat Software Inc., San Jose, CA, USA), based on n=12 images per condition (3 explants per condition, 4 images per explant). Statistical analysis between conditions has been assessed by a Student t test according Sigmaplot recommendation, with threshold of significance set to 5% ($p<0.05$).

EFFICACY

Comparative clinical study *vs* a benchmark

In vivo, two clinical studies demonstrated the cosmetic benefits of SkinNexus Collag3n. Compared to a leading benchmark, the first clinical study showed superior efficacy on wrinkle reduction and skin sagging improvement. These results were further substantiated in a second placebo-controlled clinical study (see the next chapter) further confirming rapid and sustained benefits on wrinkles and skin tonicity.

in vivo

OBJECTIVE

As skin ages, it undergoes multiple structural and functional changes driven by both intrinsic and extrinsic factors. Among these, progressive damages and loss of dermal collagen play a central role. Intrinsic aging is associated with a natural decline in collagen synthesis and increased collagen fragmentation, while extrinsic factors, most notably UV exposure, further accelerate collagen fiber degradation [Tobin, 2017; He *et al*, 2023; Quan *et al*, 2015; Shao *et al*, 2019]. On facial skin, these combined processes contribute to structural weakening, leading to visible signs of aging such as skin sagging and wrinkle formation.

In this context, the emerging concept of collagen banking aims to preserve the skin's collagen capital over time for cosmetic benefits on skin appearance, resilience and youthfulness.

To explore this approach clinically, a randomized, split-face, double-blind, comparative clinical study between SkinNexus Collag3n and a market-leading benchmark was conducted on 27 volunteers having visible crow's feet wrinkles.

SkinNexus Collag3n at 1% equivalent to 50 ppm of recombinant collagen III was compared with a market-leading recombinant collagen III benchmark (Benchmark T1, INCI Sr-(sh-polypeptide 69-dipeptide 19)), tested at a ten-fold higher dose (i.e. 500 ppm collagen III). Both products were tested in the same gel cream formulation.

Study outcomes focused on changes in jaw sagging as well as in crow's feet wrinkle volume (Fig. 10).

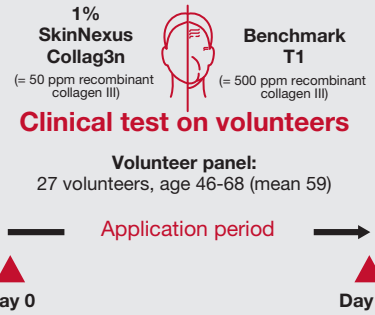


Figure 10 - Clinical study design.

RESULTS & DISCUSSION

Effect on sagging

After 28 days of application, SkinNexus Collag3n at 1% showed a significant decrease ($p<0.001$) of 19% in the sagging parameter Positive Volume compared to baseline, whereas the benchmark T1 did not demonstrate any statistically significant effect.

Overall, these results indicate a 3-fold greater efficacy compared with the benchmark T1 ($p<0.05$), despite SkinNexus Collag3n being evaluated at a 10-fold lower concentration of recombinant collagen III. This shows a higher efficacy on skin sagging after 28 days of application.

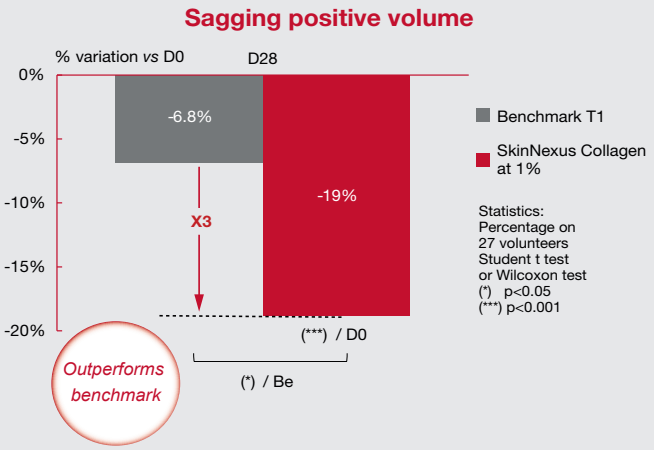


Figure 11 - Change in Sagging (Positive Volume) Over Time. SkinNexus Collag3n at 1% equivalent to 50 ppm of recombinant Col III *vs* Benchmark recombinant Col III T1 at 500 ppm.

in vivo

Representative VISIA images illustrating a reduction in sagging are shown in Figure 12.



Figure 12 - Change in sagging illustrative images (Volunteer #26).

Change in Crow's feet wrinkles volume

After 28 days of application, SkinNexus Collag3n at 1% showed a significant decrease ($p<0.05$) of 18% in Crow's Feet Wrinkle Volume as compared to baseline, while the benchmark T1 did not show a significant change in this parameter (Fig. 13).

These results indicate that SkinNexus Collag3n effectively reduces Crow's Feet Wrinkle Volume after 28 days, with an approximately 2-fold greater reduction compared to the benchmark T1.

Representative VISIA images illustrating a reduction in crow's feet wrinkles are shown in Figure 14. Green color mapped wrinkle volume visualization enables a clearer representation of wrinkle changes, thereby highlighting the efficacy of SkinNexus Collag3n.

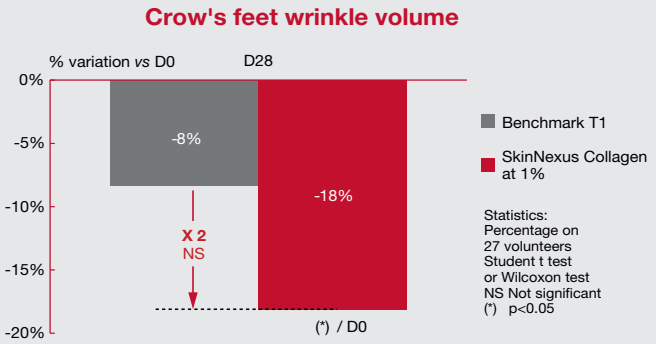


Figure 13 - Change in Volume of Crow's Feet Wrinkles Over Time. SkinNexus Collag3n at 1% equivalent to 50 ppm of recombinant Col III versus Benchmark recombinant Col III T1 at 500 ppm.

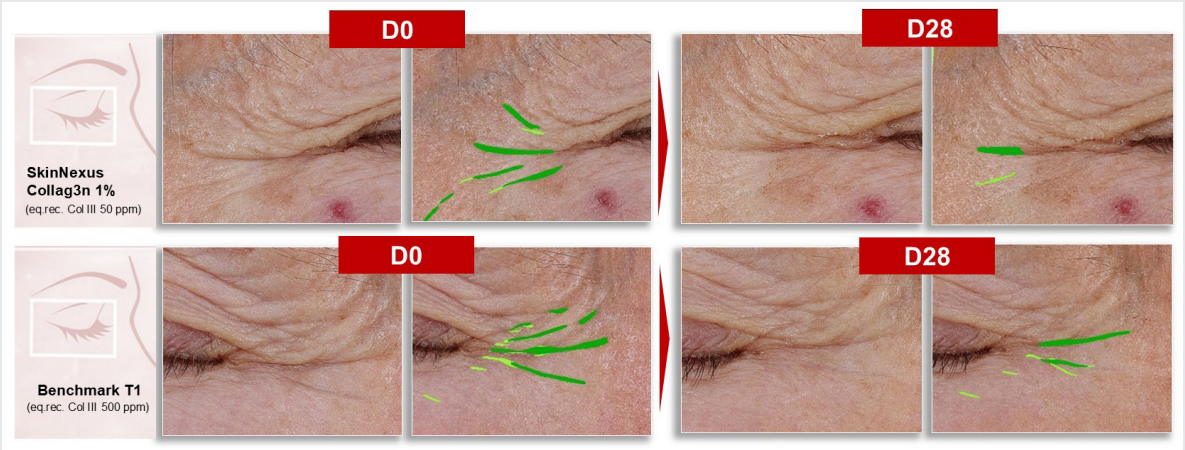


Figure 14 - Crow's Feet Wrinkle Illustrative Images (Volunteer #29) with Wrinkle Volume visualization in green at each timepoint.

CONCLUSION

SkinNexus Collag3n at 1% demonstrated the ability to improve visible signs of facial aging, including jaw sagging and crow's feet wrinkle volume, after only 28 days of application.

Overall, SkinNexus Collag3n shows clinically proven anti-aging efficacy after 28 days, outperforming a market-leading recombinant collagen benchmark, despite being evaluated at a 10-fold lower concentration.

MATERIALS&METHODS

Study number
EMB201201- 10143B

Tested products
- SkinNexus Collag3n at 1% (BC10239 batch PA241114.01) corresponding to 50 ppm (=0.005%) of recombinant collagen III.
- Benchmark T1 at 0.05% corresponding to 500 ppm of recombinant collagen III (INCI: Sr-(sh-Polypeptide 69-Dipeptide 19)).

The concentration of recombinant collagen III in the benchmark T1 formulation was 10-fold higher than that in the SkinNexus Collag3n formulation.

Study design
The comparative clinical study was conducted as a double-blind, randomized, split-face trial involving two products formulated in the same gel cream formulation as detailed in Annex 2. The efficacy of the formulations was compared to the baseline (before treatment, D0) and between them. The study was conducted during a period of 28 days with check points at D0 and D28.

Inclusion criteria
The study was done on 27 healthy female volunteers with Fitzpatrick skin type I-IV, aged from 46 to 68 years old, having visible crow's feet wrinkles.

Application modality
Volunteers applied the assigned formulations twice daily, once in the morning and once in the evening, for 28 days to one half of the face under normal conditions of use. The last application was done the evening before the D28 visit.

Evaluation methods
Jaw Sagging and crow's feet wrinkle volume were evaluated using fringe projection (AEVA)
The AVEA is a 3D acquisition system used for imaging. It combines a projection unit with stereo imaging techniques, allowing a wide range of studies and measurements, from wrinkle reduction to body reshaping. It is designed to quantify efficacy for cosmetic, aesthetic, and dermatological products and treatments.
Studied area: Crow's feet and jaw/jowl
Studied parameters: Jaw sagging positive volume and crow's feet wrinkle volume.
Lower values of both parameters reflect an improvement.

Representative images by VISIA imaging
The VISIA has a multi-point positioning system and ghost imaging capability that captures well registered images to document progress over time. The VISIA captures and automatically analyzes left, right, and frontal facial views. The multi-spectral lighting includes IntelliFlash, cross-polarized to record the surface skin conditions. VISIA images were taken to be used as representative images. In addition, Color mapping was automatically generated by the software, to illustrate the volume of the crow's feet wrinkles. The line shows the wrinkle area, with a more intense color corresponding to a thicker wrinkle.

Results and Statistics
Results are expressed as the mean percentage of change compared to the baseline measurement (D0). The comparisons between products are expressed as difference between their percentage of changes over baseline.

All data have been analysed to confirm the normality of the distribution using the Shapiro-Wilk Test. The statistical analysis of the change in the studied parameters for each product over time, as well as differences in the studied parameters between the treatment groups were performed as follows:
- Validation of the normality of the parameters studied: Paired t test.
- Invalidation of the normality of the parameters studied: Wilcoxon Signed Rank Test.

All statistical tests were run at 95% confidence level (p-value < 0.05).

EFFICACY

Placebo-controlled clinical study

in vivo

OBJECTIVE

In addition to the comparative study versus the market-leading benchmark (reported in the previous chapter), a placebo-controlled study SkinNexus Collag3n at 1%, equivalent to 50 ppm of recombinant collagen III, was conducted. It was a randomized, split-face, double-blind study on 31 volunteers having visible crow's feet wrinkles. Clinical endpoints focused on changes in skin tonicity on the jaw as well as in crow's feet wrinkles (Fig. 15).

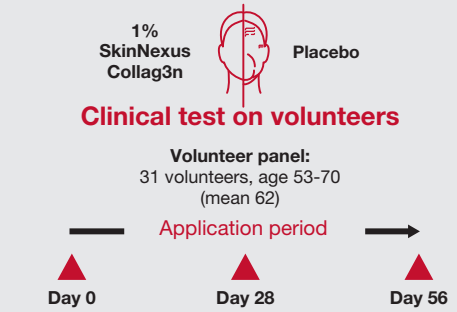


Figure 15 - Clinical study design.

RESULTS & DISCUSSION

Effect on skin tonicity

After 28 days of application, SkinNexus Collag3n at 1% showed significant decrease in R6 cutometer parameter of 28% as compared to baseline ($p<0.001$) and 9% as compared to placebo ($p<0.01$). After 56 days of application, SkinNexus Collag3n at 1% showed significant decrease in R6 of 31% as compared to baseline ($p<0.001$), and 7.5% as compared to placebo ($p<0.05$) (Fig. 16).

These results show the ability of SkinNexus Collag3n at 1% to improve skin tonicity of the jaw starting after 28 days of application with sustained effects at 56 days. It completes the anti-sagging results on the jaw obtained in the previous comparative study *vs* the benchmark.

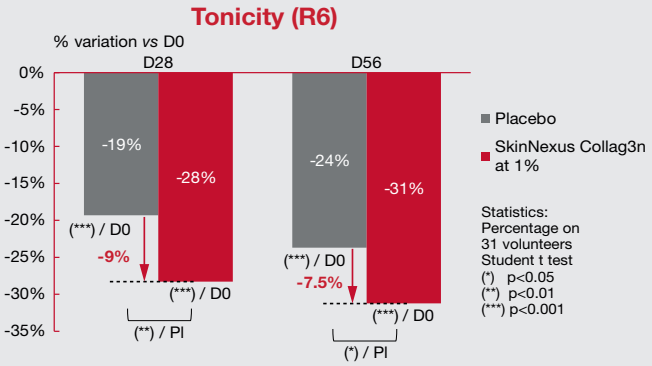


Figure 16 - Change in skin tonicity (R6) on the jaw over time.

Change in Crow's feet wrinkles roughness

SkinNexus Collag3n at 1% showed improvements in the Average Roughness (Rq) of the skin in the crow's feet area. As compared to baseline, there were significant decreases ($p<0.05$) of 6.9% after 28 days of application, and of 7.7% after 56 days of application. As compared to the placebo, SkinNexus Collag3n at 1% showed a significant decrease ($p<0.05$) of 5.5% after 56 days of application (Fig. 17).

These results confirm the smoothing effect of SkinNexus Collag3n at 1% on crow's feet wrinkles as already shown in the comparative study with the benchmark.

Representative VISIA images illustrating a reduction in crow's feet wrinkles are shown in Figure 18.

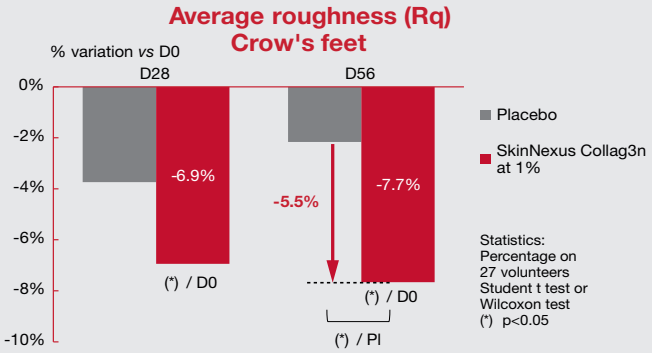


Figure 17 - Change in average roughness (Rq) of crow's feet over time.



Figure 18 - Illustrative Images of Crow's Feet Area Over Time (Volunteer #21).

in vivo

CONCLUSION

SkinNexus Collag3n at 1% demonstrated improvements of visible signs of facial aging compared to placebo, with significant benefits on skin tonicity of the jaw area and on crow's feet wrinkles.

SkinNexus Collag3n, supported by both a comparative study against a benchmark and a placebo-controlled study, demonstrates its relevance as an active ingredient for collagen banking, with clinically proven and visible benefits after 28 days on lower-face sagging and the appearance of crow's feet wrinkles.

MATERIALS&METHODS

Study number

EMB201201- 10143A

Tested products

SkinNexus Collag3n at 1% (BC10239 batch PA241114.01), corresponding to 50 ppm of recombinant collagen III.

Study design

The clinical study was conducted as a double-blind, randomized, split-face trial involving SkinNexus Collag3n at 1% in a gel cream formulation and a placebo gel cream, as detailed in Annex 3. The efficacy of the formulations was compared to the baseline (before treatment, D0) and between them. The study was conducted during a period of 56 days with check points at D0, D28 and D56.

Inclusion criteria

The study was done on 31 healthy female volunteers with Fitzpatrick skin type I-IV, aged from 53 to 70 years old, having visible crow's feet wrinkles. Only 27 volunteers completed the study on the average roughness parameter (Rq) of crow's feet.

Application modality

Volunteers applied the assigned formulations twice daily, once in the morning and once in the evening, for 56 days to one half of the face under normal conditions of use. The last application was done the evening before the visit.

Evaluation methods

Measurement of skin tonicity by cutometer

The Cutometer dual MPA 580 is used to measure the biomechanical properties of the upper skin layer. By using negative pressure, it mechanically deforms the skin. The measuring principal is based on the suction method. Negative pressure is created in the device, and the skin is drawn into the aperture of the probe and released again after a defined time. Inside the probe, the penetration depth is determined by a non-contact optical measuring system. This optical measuring system consists of a light source and a light receptor, as well as two prisms facing each other. The light intensity varies according to the penetration depth of the skin. The resistance of the skin to the negative pressure and its ability to return into its original position are displayed in real time during measurement as curves of penetration depth in distance (mm) *vs*. time (s). This allows objective quantification of the biomechanical properties of skin.

Studied area: Jaw

Studied parameter: R6 viscoelastic parameter (Uv/Ue, Viscoelasticity index, ratio of delayed-to-immediate distension). It is increased with age, and a decrease in this parameter suggests an improvement in skin tonicity.

Measurement of crow's feet roughness using fringe projection (AEVA)

The AVEA is a 3D acquisition system for imaging. It has a projection unit combined with stereo imaging techniques that allow it to address wrinkle reduction. It is designed to quantify efficacy assessment for cosmetics, aesthetical and dermatology products and treatments.

Studied area: Crow's feet

Studied parameter: Rq, average roughness. A decrease of this parameter indicates an improvement of skin surface smoothness.

Representative images by VISIA imaging

The VISIA has a multi-point positioning system and ghost imaging capability that captures well registered images to document progress over time. The VISIA captures and automatically analyzes left, right, and frontal facial views. The multi-spectral lighting includes IntelliFlash, cross-polarized and UV lighting to record the skin surface conditions. VISIA images were taken to be used as representative images.

Results and Statistics

Results are expressed as the mean percentage of change compared to the baseline measurement (D0). The comparisons between products are expressed as difference between their percentage of changes over baseline.

All data have been analysed to confirm the normality of the distribution using the Shapiro-Wilk Test. The statistical analysis of the change in the studied parameters for each product over time, as well as differences in the studied parameters between the treatment groups were performed as follows:

- Validation of the normality of the parameters studied: Paired t test.
- Invalidation of the normality of the parameters studied: Wilcoxon Signed Rank Test.

All statistical tests were run at 95% confidence level (p-value < 0.05).



GENERAL CONCLUSION

SkinNexus Collag3n: advancing the future of collagen banking

Supporting the skin's collagen capital to sustain long-term structural youth and visible anti aging performance.

- 100% Human-identical vegan recombinant collagen III.
- Biologically active, supporting the synthesis of collagens within dermal matrix and epidermal regeneration.
- Clinically proven collagen repair and anti-aging performance, outperforming a leading peptide benchmark, at a 10-fold lower dose.

Among the multiple factors underlying skin quality and perceived youthfulness, the preservation of the dermal collagen network plays a decisive role. From early adulthood, collagen synthesis progressively declines while collagen degradation accelerates, leading to a gradual loss of dermal density and regenerative capacity.

Built on the collagen banking concept, SkinNexus Collag3n introduces a next generation recombinant collagen III solution designed to proactively preserve the skin's collagen capital, which progressively declines with age.

Fully human-identical and produced using AI-driven design and advanced biotechnology, SkinNexus Collag3n is a biologically potent cosmetic ingredient while overcoming the limitations of animal derived and first generation recombinant collagens in terms of reproducibility, purity and scalability.

In vitro, SkinNexus Collag3n promotes fibroblast proliferation and strongly stimulates the synthesis of collagen I, III and V supporting dermal structure and resilience. At the epidermal level, it enhances keratinocyte adhesion, proliferation, and epidermis regeneration.

In vivo, two clinical studies, on 27 and 31 volunteers, demonstrated the anti-aging cosmetic benefits of SkinNexus Collag3n when applied at 1%. Across these two clinical studies against benchmark and placebo, after 28 days, it significantly improved jaw sagging by 19%, jaw tonicity by 28% and wrinkle volume by 18%. Notably, SkinNexus Collag3n outperformed a leading market recombinant collagen benchmark despite being evaluated at a 10-fold lower concentration.

SkinNexus Collag3n represents a new standard for collagen banking, combining Human identical recombinant collagen III, strong biological relevance and clinically validated performance.

It offers a high-value, science-backed cosmetic bioactive ingredient designed to proactively preserve the skin's structural youth capital and meet the growing demand for effective, long-term and accessible solutions for lower-face sagging and improved wrinkle appearance.

ANNEXES

Annex 1 - Technical data - Available upon request

- Quality and regulatory product information
- Toxicological abstract
- Composition sheet
- Specifications
- Formulation Data Sheet
- Natural content origin (according to ISO 16128)
- Product Purity Profile

Annex 2 - Clinical test gel cream formula

Comparative study *vs* benchmark

Trade name	INCI name	Benchmark %	SkinNexus Collag3n %
Water	Water	qsf. 100	qsf. 100
Cetiol C 5C	Coco-Caprylate/Caprate	6.50	6.50
Dimethyl Isosorbide	Dimethyl Isosorbide	2.00	2.00
Glycerine 99.5%	Glycerin	2.00	2.00
Dehymuls PGPH	Polyglyceryl-2 Dipolyhydroxy-stearate	2.00	2.00
Eumulgin SML 20	Polysorbate 20	2.00	2.00
Plantapon LGC Sorb	Sodium Lauryl Glucose Carboxylate, Lauryl Glucoside	1.75	1.75
Cosmedia SP	Sodium Polyacrylate	1.50	1.50
Euxyl PE 9010	Phenoxyethanol, Ethylhexylglycerin	1.10	1.10
Verdessence Xanthan	Xanthan Gum	0.10	0.10
Eumulgin Prisma	Disodium Cetearyl Sulfosuccinate	0.08	0.08
SkinNexus Collag3n C or SkinNexus Collag3n P	Mannitol (and) Collagen or Mannitol (and) sh-Polypeptide-123	-	1*
Benchmark T1	Sr-(sh polypeptide 69- dipeptide 19)	0.05	-

* Corresponding to 0.005% Collagen / sh-Polypeptide-123

Clinical test gel cream formula

Comparative study *vs* placebo

Trade name	INCI name	Benchmark %	SkinNexus Collag3n %
Water	Water	qsf. 100	qsf. 100
Cetiol C 5C	Coco-Caprylate/Caprate	6.50	6.50
Dimethyl Isosorbide	Dimethyl Isosorbide	2.00	2.00
Glycerine 99.5%	Glycerin	2.00	2.00
Dehymuls PGPH	Polyglyceryl-2 Dipolyhydroxy-stearate	2.00	2.00
Eumulgin SML 20	Polysorbate 20	2.00	2.00
Plantapon LGC Sorb	Sodium Lauryl Glucose Carboxylate, Lauryl Glucoside	1.75	1.75
Cosmedia SP	Sodium Polyacrylate	1.50	1.50
Euxyl PE 9010	Phenoxyethanol, Ethylhexylglycerin	1.10	1.10
Verdessence Xanthan	Xanthan Gum	0.10	0.10
Eumulgin Prisma	Disodium Cetearyl Sulfosuccinate	0.08	0.08
SkinNexus Collag3n C or SkinNexus Collag3n P	Mannitol (and) Collagen or Mannitol (and) sh-Polypeptide-123	-	1*

* Corresponding to 0.005% Collagen / sh-Polypeptide-123

BIBLIOGRAPHY

Bai M, Kang N, Xu Y, Wang J, Shuai X, Liu C, Jiang Y, Du Y, Gong P, Lin H, Zhang X.
The influence of tag sequence on recombinant humanized collagen (rhCol) and the evaluation of rhCol on Schwann cell behaviors.
Regen Biomater. 2023 Oct 16;10:rbad089.

Cheng W, Yan-Hua R, Fang-Gang N, Gua-an Z.
The content and ratio of type I and III collagen in skin differ with age and injury.
African Journal of Biotechnology Vol. 10(13), pp. 2524-2529, March, 2011.

Choudhary V, Choudhary M, Bollag W.B.
Exploring Skin Wound Healing Models and the Impact of Natural Lipids on the Healing Process.
Int. J. Mol. Sci. 2024, 25, 3790.

Davidenko N, Schuster CF, Bax DV, Farndale RW, Hamaia S, Best SM, Cameron RE.
Evaluation of cell binding to collagen and gelatin: a study of the effect of 2D and 3D architecture and surface chemistry.
J Mater Sci Mater Med. 2016 Oct;27(10):148.

Davis GE. Affinity of integrins for damaged extracellular matrix: $\alpha(v)\beta(3)$ binds to denatured collagen type I through RGD sites. Biochem Biophys Res Commun. 1992 Feb 14;182(3):1025-31.

Emsley J, Knight CG, Farndale RW, Barnes MJ, Liddington RC.
Structural basis of collagen recognition by integrin $\alpha2\beta1$.
Cell. 2000 Mar 31;101(1):47-56.

Fisher GJ, Varani J, Voorhees JJ.
Looking older: fibroblast collapse and therapeutic implications.
Arch Dermatol. 2008 May;144(5):666-72.

Fujisaki H, Futaki S, Mizuno K, Hattori S.
Evaluation of Keratinocyte Proliferation on Two- and Three-dimensional Type I Collagen Substrates.
J Vis Exp. 2019 Apr 22;(146).

Gao J, Guo Z, Zhang Y, Liu Y, Xing F, Wang J, Luo X, Kong Y, Zhang G.
Age-related changes in the ratio of Type I/III collagen and fibril diameter in mouse skin.
Regen Biomater. 2022 Dec 29;10: rbac110.

Guo X, Ma Y, Wang H, Yin H, Shi X, Chen Y, Gao G, Sun L, Wang J, Wang Y, Fan D.
Status and developmental trends in recombinant collagen preparation technology.
Regen Biomater. 2023 Nov 29;11:rbad106.

Hamaia S, Farndale RW.
Integrin recognition motifs in the human collagens.
In: Gullberg D, editor. I Domain Integrins. Adv Exp Med Biol. 2014;819:127 142.

He C, Wang R, Zhang Q, Wang Y, Huang N.
In Vitro Bioactivity of a Recombinant Human Collagen Peptide in a Filler Biomimetic Skin Model.
J Cosmet Dermatol. 2025 Dec;24(12):

He T, Fisher GJ, Kim AJ, Quan T.
Age-related changes in dermal collagen physical properties in human skin.
PLoS One. 2023 Dec 8;18(12).

Jadach B, Mielcarek Z, Osmałek T.
Use of Collagen in Cosmetic Products.
Curr Issues Mol Biol. 2024 Mar 4;46(3):2043-2070.

Jie IWK, Lee KWA, Yoon SE, Song JK, Chan LKW, Lee CH, Jeong E, Kim JH, Yi KH.
Advancements in Clinical Utilization of Recombinant Human Collagen: An Extensive Review.
Life (Basel). 2025 Apr 1;15(4):582.

Kanniyappan H, Chathurika Rathnayake RA, Osamor J, Islam M, Wang RR. The role of collagen and collagen I/III ratio in pathological conditions: insights into molecular mechanisms and therapeutic approaches. Front Bioeng Biotechnol. 2025 Oct 9; 13:1679625.

Kim YU, Gi H, Jeong EK, Han S, Seo WY, Kim YJ, Lee SB, Kim K. Development of a highly effective recombinant protein from human collagen type III $\alpha1$ (COL3A1) to enhance human skin cell functionality. BMB Rep. 2024 Sep;57(9):424-429.

Knight CG, Morton LF, Peachey AR, Tuckwell DS, Farndale RW, Barnes MJ.
The collagen-binding A-domains of integrins $\alpha(1)\beta(1)$ and $\alpha(2)\beta(1)$ recognize the same specific amino acid sequence, GFOGER, in native (triple-helical) collagens.
J Biol Chem. 2000 Jan 7;275(1):35-40.

Kuivaniemi H, Tromp G.
Type III collagen (COL3A1): Gene and protein structure, tissue distribution, and associated diseases.
Gene. 2019 Jul 30; 707:151-171.

Lynn AK, Yannas IV, Bonfield W.
Antigenicity and immunogenicity of collagen.
J Biomed Mater Res B Appl Biomater. 2004 Nov 15;71(2):343-54.

Mathew-Steiner SS, Roy S, Sen CK.
Collagen in Wound Healing.
Bioengineering (Basel). 2021 May 11;8(5):63.

Naharro-Rodriguez J, Bacci S, Hernandez-Bule ML, Perez-Gonzalez A, Fernandez-Guarino M.
Decoding Skin Aging: A Review of Mechanisms, Markers, and Modern Therapies.
Cosmetics 2025, 12, 144.

Oikarinen A.
Aging of the skin connective tissue: how to measure the biochemical and mechanical properties of aging dermis.
Photodermatol Photoimmunol Photomed. 1994 Apr;10(2):47-52.

Quan T, Fisher GJ.
Role of Age-Associated Alterations of the Dermal Extracellular Matrix Microenvironment in Human Skin Aging: A Mini-Review.
Gerontology. 2015;61(5):427-34.

Quan T.
Molecular insights of human skin epidermal and dermal aging.
J Dermatol Sci. 2023 Nov;112(2):48-53.

Reilly DM, Lozano J.
Skin collagen through the lifestages: importance for skin health and beauty.
Plast Aesthet Res 2021;8:2.

Salamito M, Haydont V, Pageon H, Ruggiero F, Girardeau-Hubert S.
Collagen diversity in human skin: Aging, wound healing, and disorders.
Matrix Biol. 2025 Sep; 140:133-153.

Shao Y, Qin Z, Alexander Wilks J, Balimunkwe RM, Fisher GJ, Voorhees JJ, Quan T.
Physical properties of the photodamaged human skin dermis: Rougher collagen surface and stiffer/harder mechanical properties.
Exp Dermatol. 2019 Aug;28(8):914-921.

Siljander PR, Hamaia S, Peachey AR, Slatter DA, Smethurst PA, Ouwehand WH, Knight CG, Farndale RW.
Integrin activation state determines selectivity for novel recognition sites in fibrillar collagens.
J Biol Chem. 2004 Nov 12;279(46):47763-72.

Singh D, Rai V, Agrawal DK.
Regulation of Collagen I and Collagen III in Tissue Injury and Regeneration.
Cardiol Cardiovasc Med. 2023;7(1):5-16.

Smith LT, Holbrook KA, Madri JA.
Collagen types I, III, and V in human embryonic and foetal skin.
Am J Anat. 1986 Apr;175(4):507-21.

Sun M, Chen S, Adams SM, Florer JB, Liu H, Kao WW, Wenstrup RJ, Birk DE.
Collagen V is a dominant regulator of collagen fibrillogenesis: dysfunctional regulation of structure and function in a corneal-stroma-specific Col5a1-null mouse model.
J Cell Sci. 2011 Dec 1;124(Pt 23):4096-105.

Taubenberger AV, Woodruff MA, Bai H, Muller DJ, Hutmacher DW.
The effect of unlocking RGD-motifs in collagen I on pre-osteoblast adhesion and differentiation.
Biomaterials. 2010 Apr;31(10):2827-35.

Tobin DJ.
Introduction to skin aging.
J Tissue Viability. 2017 Feb;26(1):37-46.

Varani J, Dame MK, Rittie L, Fligiel SE, Kang S, Fisher GJ, Voorhees JJ.
Decreased collagen production in chronologically aged skin: roles of age-dependent alteration in fibroblast function and defective mechanical stimulation.
Am J Pathol. 2006 Jun;168(6):1861-8.

Wang Y, Cai W, Kong K, Jia X, Anderson RR, Franco W.
Visualization of neo-epidermis formation and evaluation of wound closure using UV fluorescence excitation imaging at two wavelengths (Conference Presentation).
Proc. SPIE 10851, Photonics in Dermatology and Plastic Surgery 2019, 108510H (4 March 2019).

Wang Y, Kuang S, Fan Y, Li D, Li L, Wang J, Bian W.
Structural characterization of recombinant humanized type III collagen cold gel and its effects on the biological functions of HaCaT cells.
Front Med (Lausanne). 2026 Feb 10; 13:1749460.

Wenstrup RJ, Florer JB, Brunskill EW, Bell SM, Chervoneva I, Birk DE.
Type V collagen controls the initiation of collagen fibril assembly.
J Biol Chem. 2004 Dec 17;279(51):53331-7.

Wong CW, LeGrand CF, Kinnear BF, Denil SLIJ, Dye DE, Benny P, Chan JKY, Lane EB, Coombe DR.
Keratinocyte self-renewal and differentiation is dictated by extrinsic signals from dermal extracellular matrices.
Open Biol. 2025 Oct;15(10):240417.

Zhang J, Yu H, Man MQ, Hu L.
Aging in the dermis: Fibroblast senescence and its significance.
Aging Cell. 2024 Feb;23(2):e14054.

Zorina A, Zorin V, Kudlay D, Kopnin P.
Age-Related Changes in the Fibroblastic Differon of the Dermis: Role in Skin Aging.
Int J Mol Sci. 2022 May 30;23(11):6135.

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